Operation Instruction

(Translation of original instructions)

C- Pro 2.0 Controller
(FS&RS Firmware)

1.1 ELECTRICAL PERFORMANCE

Power supply: AC24V (-5%/+15%), 30W.

Ambient temperature: 0-50°C

1.2 OPERATOR PANEL

1 Start: Start the air compressor on the main screen

2 Stop: Stop the air compressor in any case

3 Move to the left when modifying parameter; Manually loaded

4 Move to the right when modifying parameter; Manually unloaded

5 Confirm the parameter modification and enter the next interface

6 Cancel the parameter modification, reset fault and quit the modification status

7 Up and down arrow keys: Page up and down and parameters modification

8 POWER — Power indicator: It is constantly on when the controller is powered normally

9 RUN — Run indicator: It is constantly on when the air compressor is running normally

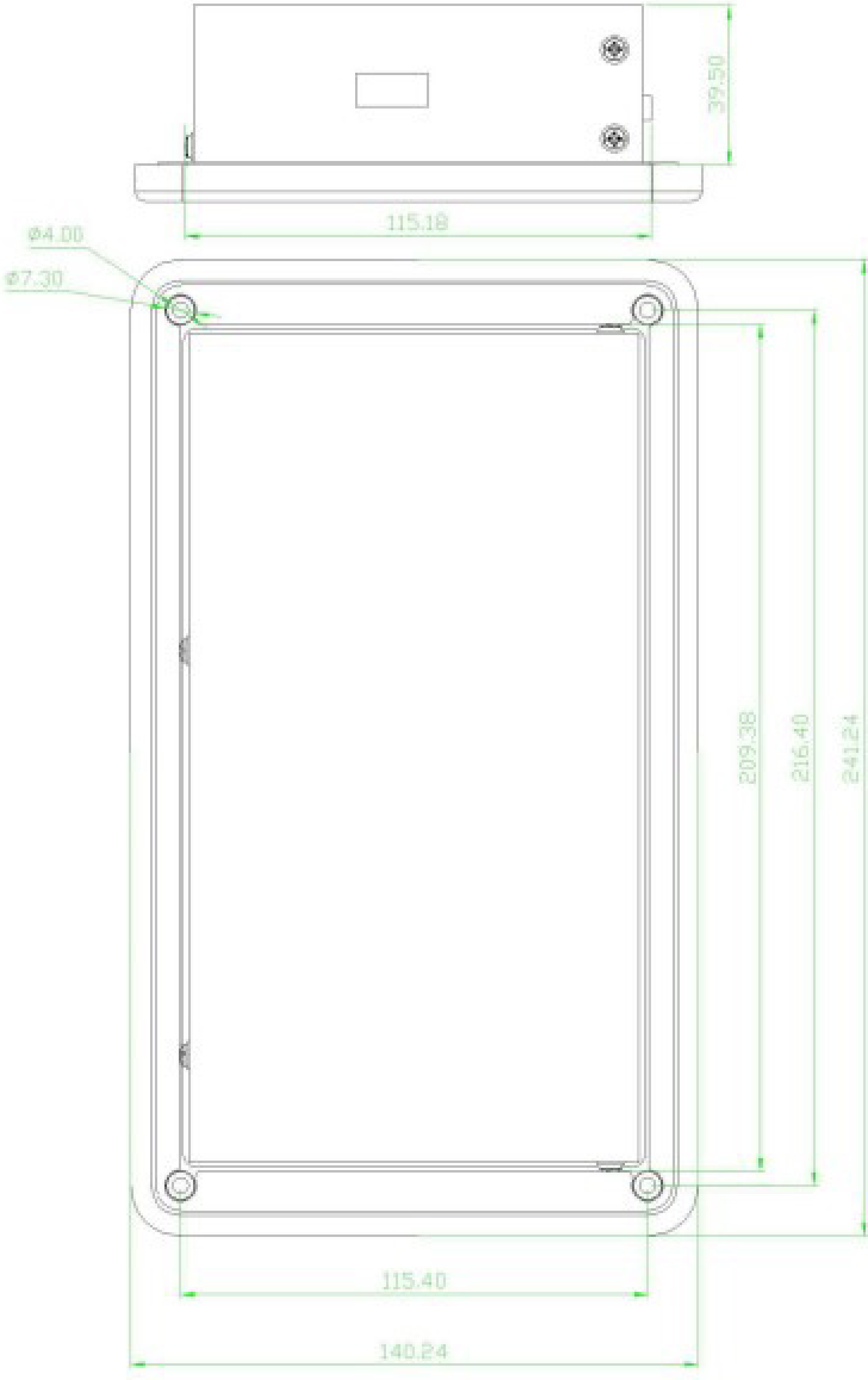
10 SERVICE — Service indicator: It is constantly on when the air compressor needs service

11 SHUTDOWN — Shutdown indicator: It will flash in case of system warning, and will be constantly on in case of fault shutdown

Note

The above picture shows the initial screen including the following contents:
Drawing No. : “CPN: QX187453”
Software venison No. : “VER.: DG-V100”

1.3 INSTALLATION DIMENSIONS



1.4 DESCRIPTION OF TERMINALS

Digital output port 8			
Terminal	**No.**	**Function definitions**	**Remarks**
X01	C-R	Common terminal	Normally open/passive contact
	R8	Fault signal output	Normally open/passive contact
	C-R	Common terminal	Normally open/passive contact
	R7	Warning signal output	Normally open/passive contact
X02	C-R	Common terminal	Normally open/active output
	R6	Fan motor contactor signal output	Normally open/active output
	R5	Reserve	Normally open/active output
	R4	Loading valve signal output	Normally open/active output
X03	C-R	Common terminal	Normally open/active output
	R3	Star (Y) start signal output(FS)/ Reserve(RS)	Normally open/active output
	R2	Delta (Δ) start signal output(FS)/ Reserve(RS)	Normally open/active output
	R1	Main contactor signal output(FS)/ Cabinet fan output(RS)	Normally open/active output

Current transformer input port 2			
Terminal	**No.**	**Function definitions**	**Remarks**
X04 CT2	a	Fan motor current transformer access	Effective range 0~100mA
	C-CT		
X05 CT1	a	Main motor current transformer access a(FS)/ Reserve(RS)	Effective range 0~350mA
	b	Main motor current transformer access b(FS)/ Reserve(RS)	
	C	Main motor current transformer access c(FS)/ Reserve(RS)	

Power input			
Terminal	**No.**	**Function definitions**	**Remarks**
X06	24Vac	24VAC access port	
	24VaC		
	GND	Common ground	

Analog output port 1			
Terminal	**No.**	**Function definitions**	**Remarks**
X07	ABA-OUT	Analog output	4~20mA Current output
	AGND		

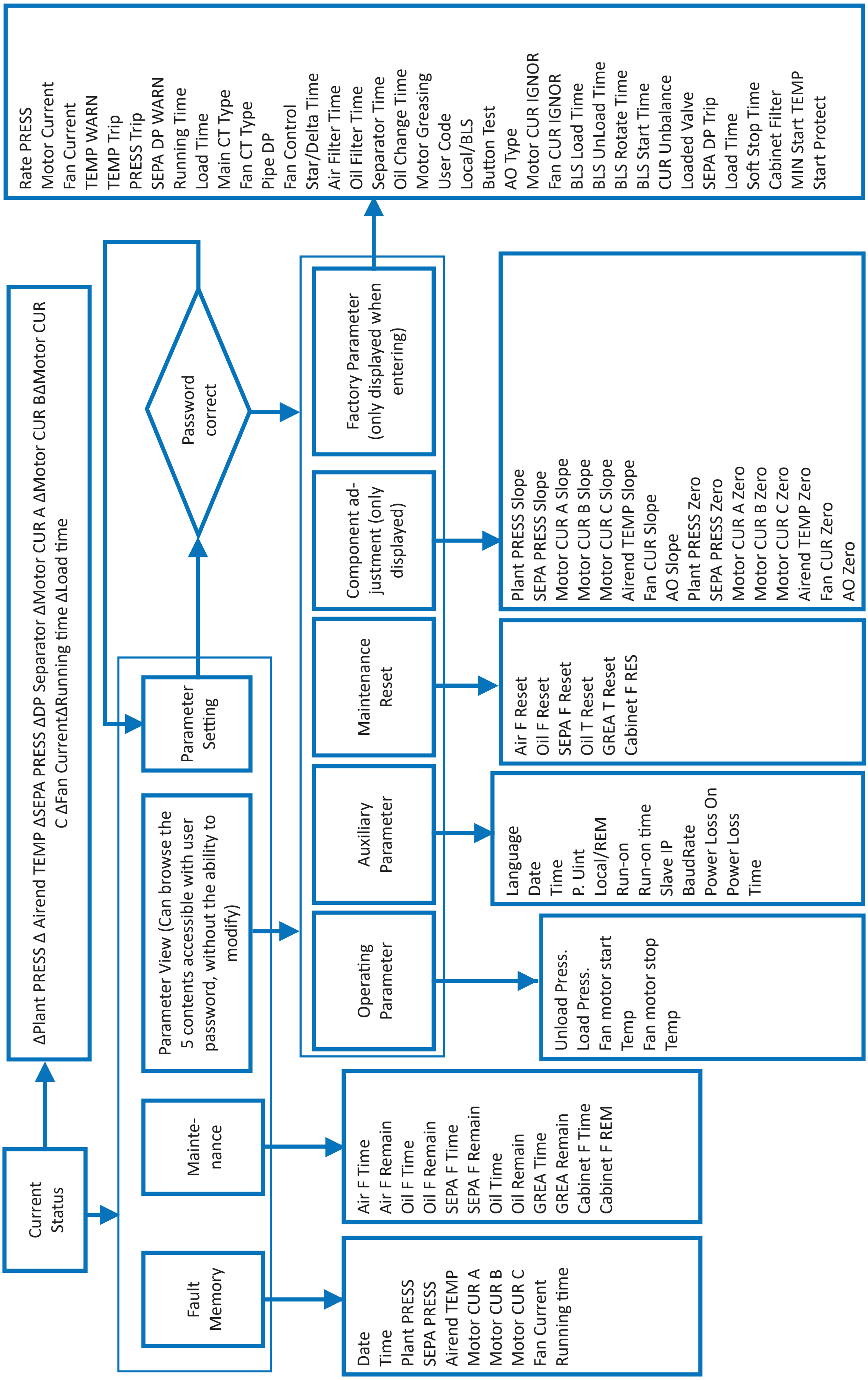
Analog input port 3			
Terminal	**No.**	**Function definitions**	**Remarks**
X08	C-ANA 1	Airend temperature T sensor terminal (+)	Pt100 platinum resistance Temperature Sensor
	AN1	Airend temperature T sensor terminal (-)	
	AGNF	AGND	
	C-ANA 2	Plant pressure P1 detection sensor terminal (+24V)	4~20mA current Pressure Sensor
	AN2	Plant pressure P1 detection sensor terminal	
	C-ANA 3	Separator pressure P2 detection sensor terminal (+24V)	4~20mA current Pressure Sensor
	AN3	Separator pressure P2 detection sensor terminal	

Digital input port 8			
Terminal	**No.**	**Function definitions**	**Remarks**
X09	C-DI	Common terminal	
	DI1	Emergency stop	Normally closed/passive contact
	DI2	DI remote start/stop	Normally open/passive contact
	DI3	Fan motor feedback	Normally closed/passive contact
	DI4	Air filter DP	Normally closed/passive contact
	DI5	Remote start/stop enable	Normally open/passive contact
	DI6	Remote load/unload	Normally open/passive contact
	DI7	External fault	Normally open/passive contact
	DI8	Main motor temperature	PTC thermistor

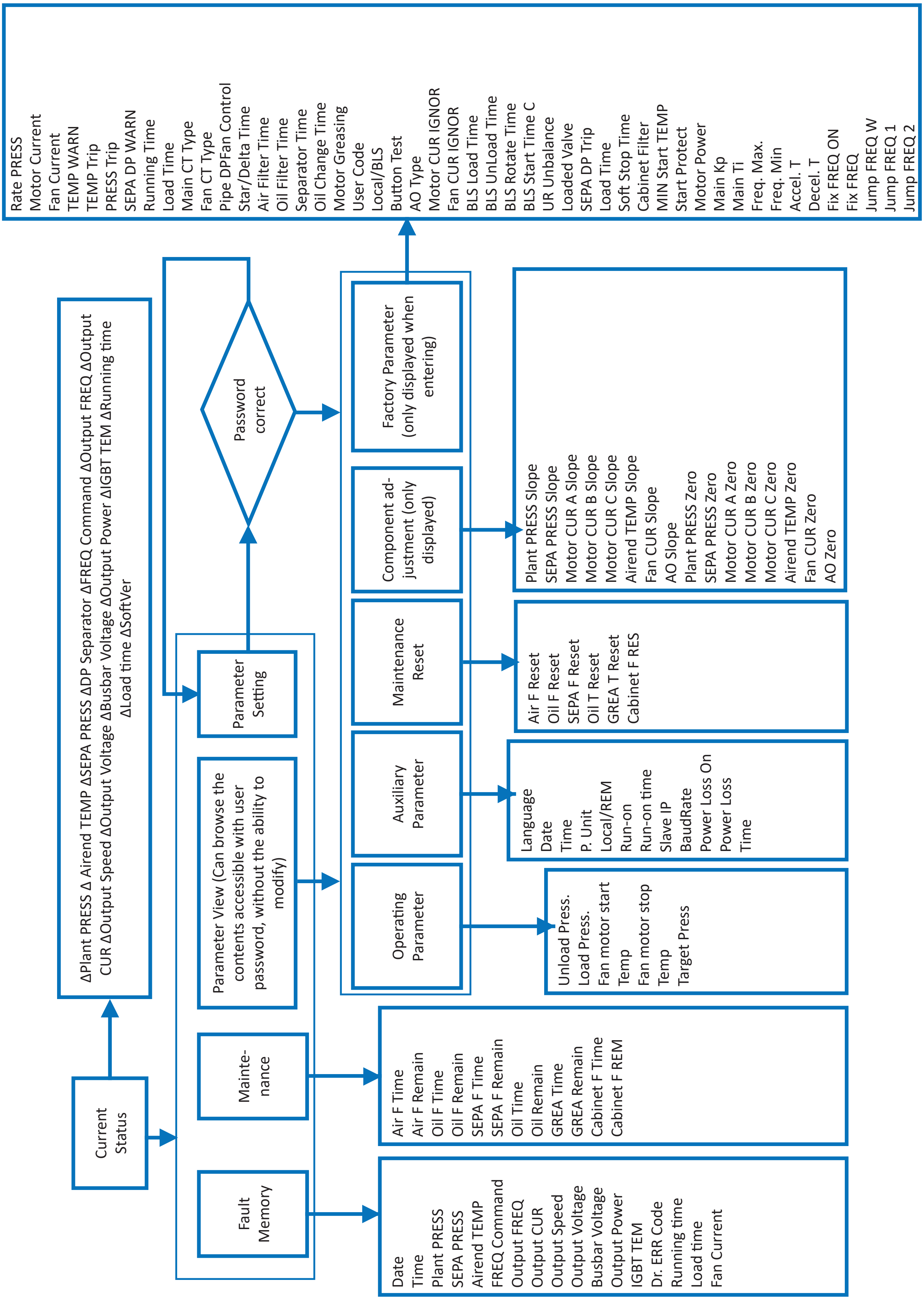
RS485 communication port 2			
Terminal	**No.**	**Function definitions**	**Remarks**
X10	L1+	RS485+	DCS monitoring communication port
	L2-	RS485-	
	GND	GND	
X11	L1+	RS485+	Invertor Drive communication port
	L2-	RS485-	
	GND	GND	

2.1 OPERATION INTERFACE FLOW CHART

FS machine



RS machine


Current Status
ΔPlant PRESS Δ Airend TEMP ΔSEPA PRESS ΔDP Separator ΔFREQ Command ΔOutput FREQ ΔOutput CUR ΔOutput Speed ΔOutput Voltage ΔBusbar Voltage ΔOutput Power ΔIGBT TEM ΔRunning time ΔLoad time ΔSoftVer
Fault Memory
Mainte-nance
Parameter View (Can browse the contents accessible with user password, without the ability to modify)
Parameter Setting
Password correct
Operating Parameter
Auxiliary Parameter
Maintenance Reset
Component ad-justment (only displayed)
Factory Parameter (only displayed when entering)
Date
Time
Plant PRESS
SEPA PRESS
Airend TEMP
FREQ Command
Output FREQ
Output CUR
Output Speed
Output Voltage
Busbar Voltage
Output Power
IGBT TEM
Dr. ERR Code
Running time
Load time
Fan Current
Air F Time
Air F Remain
Oil F Time
Oil F Remain
SEPA F Time
SEPA F Remain
Oil Time
Oil Remain
GREA Time
GREA Remain
Cabinet F Time
Cabinet F REM
Unload Press.
Load Press.
Fan motor start
Temp
Fan motor stop
Temp
Target Press
Language
Date
Time
P. Unit
Local/REM
Run-on
Run-on time
Slave IP
BaudRate
Power Loss On
Power Loss
Time
Air F Reset
Oil F Reset
SEPA F Reset
Oil T Reset
GREA T Reset
Cabinet F RES
Plant PRESS Slope
SEPA PRESS Slope
Motor CUR A Slope
Motor CUR B Slope
Motor CUR C Slope
Airend TEMP Slope
Fan CUR Slope
AO Slope
Plant PRESS Zero
SEPA PRESS Zero
Motor CUR A Zero
Motor CUR B Zero
Motor CUR C Zero
Airend TEMP Zero
Fan CUR Zero
AO Zero
Rate PRESS
Motor Current
Fan Current
TEMP WARN
TEMP Trip
PRESS Trip
SEPA DP WARN
Running Time
Load Time
Main CT Type
Fan CT Type
Pipe DPFan Control
Star/Delta Time
Air Filter Time
Oil Filter Time
Separator Time
Oil Change Time
Motor Greasing
User Code
Local/BLS
Button Test
AO Type
Motor CUR IGNOR
Fan CUR IGNOR
BLS Load Time
BLS UnLoad Time
BLS Rotate Time C
BLS Start Time C
UR Unbalance
Loaded Valve
SEPA DP Trip
Load Time
Soft Stop Time
Cabinet Filter
MIN Start TEMP
Start Protect
Motor Power
Main Kp
Main Ti
Freq. Max.
Freq. Min
Accel. T
Decel. T
Fix FREQ ON
Fix FREQ
Jump FREQ W
Jump FREQ 1
Jump FREQ 2

2.2 INITIALIZATION INTERFACE

After the controller is powered on. The power indicator on the operation panel will be ON and the system enters the initial screen (Fig. 2.2.1). After 5s the default interface of the system will display. The initial screen shows controller part number and software version number.

CPN: QX187453
VER.: V100

Fig.2.2.1

2.3 REAL-TIME STATUS

The user interface display is divided into two parts: the upper part and the lower part. The upper part always displays the compressor's operating status. And the lower part displays the current status values and parameter setting of the compressor.

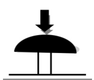
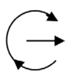
The upper part includes: warning sign, plant pressure, running conditions, real time, control mode, and Power-off Restart functional setting, Load %(RS)

	0.0 Bar
Change Air Filter	
LOCAL Ctr	Auto Restart
Current status	
Plant PRESS	---Bar
Bar Airend TEMP	---°C
SEPA PRESS	---Bar
DP Separator	---Bar

Fig.2.3.1

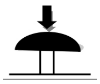
Running status include: ready to start/SD start/load/unload/manually unload/manually load/delayed stop/stop unloaded/IDEL/warning and fault message.

Warning signs and icons of running status have the following meanings:

	The compressor occur an early warning with corresponding written prompt, but the compressor won't stop.
	The compressor occur a major fault with corresponding written prompt, and the compressor stops immediately. Immediate elimination is required.
	The Emergency Stop button is pressed down.
	Unloaded running
	Loaded running
	Run-on time
	Stopped

2.4 CURRENT STATUS

The default interface of the system is the "Current Status" screen, where you can press keys to scroll through the pages to check other status parameters. Please refer to section 2.1 for status data that can be displayed. Press to enter the menu page.



	0.0 Bar
LOCALE Ctr	10 : 20 : 00
Current status	
Plant PRESS	0.0 Bar
Bar Airend TEMP	2.0 °C
SEPA PRESS	0.0 Bar
DP Separator	0.0 Bar

Fig.2.4.1

2.5 MENU PAGE

On the "Current Status" page, press to enter the "Menu" page. Press keys on the controller panel to scroll through pages down and up.

0.0 Bar	
LOCALE Ctr	10 : 20 : 00
Main Menu	
Fault Memory	
Parameter View	
Maintenance	
Parameter Setting	

Fig.2.5.1

The display with black text on white background as shown in the figure2.5.1. It is a selected item. Press keys to select the corresponding item.

If an item is selected. Press to enter the corresponding item.

2.6 FAULT MEMORY

"Fault Memory" is used to display and record the system status in case of a major fault occurs. If no failure occurs presently. It will show as Fig. 2.6.1.

Main Menu
Fault Memory
Parameter View
Maintenance
Parameter Setting

Fig.2.6.1

If a fault warning is recorded. It will show as Fig. 2.6.2.

Fault Memory

Temperature sensor error

Fig.2.6.2

A maximum of 15 records can be displayed. And no more than 5 major fault messages shall be recorded during the period from the last shutdown to the next startup. If there have been more than 5 major fault messages. They shall only be displayed but not be recorded. And new fault can be recorded 5 seconds after the compressor restarts.

Press to view the status of air compressor system when the fault occurs as shown in Fig. 2.6.3.

Temperature sensor error

Date	2012-10-31
Time	01 : 01 : 10
Plant PRESS	--- Bar
SEPA PRESS	--- Bar

Fig.2.6.3

Press for more information. The information about the system status which be recorded in the fault memory please refer to section 2.1.

Press Ⓒ to return to the previous page.

2.7 PARAMETER VIEW

Menu “Parameter View” can display all user-visible parameter setting. Those parameters can only be viewed but not be modified.

2.8 MAINTENANCE

Menu “Maintenance” can display wearing parts maintenance data. Please refer to section 2.1 for details.

Maintenance

Air F time	0010	h
Air F Remain	1990	h
Oil F time	0010	h
Oil F Remain	1990	h

Fig.2.8.1

2.9 PARAMETER SETTING

Select menu “Parameter Setting” on the main menu page. And press key to open the access code input window as (Fig. 2.9.1).

There are two levels of access codes which includ user access code (default: 1111, which can be changed) and factory access code. How to input access code is the same as the modification method of numerical parameters in section 2.10.

Parameter setting

Input code: 0000

Fig.2.9.1

Input the correct access code to enter the “Parameters Setting” menu. There are 5 sub-options under the “Parameter Setting” menu: “Operating parameter”; “Auxiliary parameter”; “ Maintenance reset”; “Element calibration”; “Factory parameter” (Fig. 2.9.2). Press to enter the selected sub-option.

Parameter setting

Operating Parameter
Auxiliary parameter
Maintenance Reset
Component adjustment

Fig.2.9.2

For parameters view and modification methods please refer to section 2.10.

2.10 PARAMETERS VIEW AND MODIFICATION METHODS

To browse the parameters: Enter menu “Parameter View” of the main menu to view user parameters.

To change the parameters setting: Enter menu “Parameter Setting” and change the parameter value as follow.

There are two types of parameters.Numerical parameters and functional parameters. For example parameter “Load Press” of the menu “Operating Parameter” is numerical parameter. And parameter “DI REM” of menu “Auxiliary Parameter” is functional parameter. how to modify the two types of parameters as follow.

1 . Modification of numerical parameters

Input access code to enter the menu “Parameter Setting” and select menu “Operating Parameter”; Press to enter the control parameter setting screen (Fig. 2.10.1).

Operating Parameter

Unload PRESS	8.0	B a r
Load PRESS	7.5	Bar
Fan Start T	080	°C
Fan Stop T	070	°C

Fig.2.10.1

Then press ∧∨ to select other parameters.
Select parameter "Load PRESS" and press ↵ to enter the modifiable status.The parameter value will flash in the last digit.

Every time you press < , the flashing digit in the parameter value will move one bit to the left. While the flashing digit in the parameter value will move one bit to the right if you press > . Press ∧∨ keys on the panel to increase or decrease the flashed digit.

After finishing the parameter modification. Press ↵ to save the new parameter value and exit the modification status.
If you want to cancel your modification. Press Ⓒ key. The original parameter value will remain. And exit the modification status at the same time.

2. Modification of functional parameters

Input access code to enter the "Parameter Setting" menu and select menu "Control Parameter"; Press ↵ to enter the control parameter screen (Fig. 2.10.2).

Auxiliary Parameter	
DI REM	ON
Run-on	ON
Run-on time	100 S
Slave IP	01

Fig.2.10.2

Press ↵ and the word "ON" indicator will be flashing.
Press ∧∨ keys to change the parameter value to "ON" or "OFF" (Fig. 2.10.3, Fig. 2.10.4).

Auxiliary Parameter	
DI REM	ON
Run-on	ON
Run-on time	100 S
Slave IP	01

Fig.2.10.3

Auxiliary Parameter	
DI REM	OFF
Run-on	ON
Run-on time	100 S
Slave IP	01

Abb.2.10.4

Press " ↵ " key to change the parameter value to the value currently displayed and exit the modification mode; press Ⓒ to retain the original parameter value and exit the modification mode.

2.11 DISPLAY OF MAJOR FAULT

When a major fault occurs in the system. It will popup the major fault warning window directly (Fig. 2.11.1).

Fault
Fault EM-stop
01:52:50
2012-10-31

Fig.2.11.1

When the major fault warning window popup. It can not be exit unless the major fault cause is eliminated. After the major fault cause has been eliminated. Press Ⓒ to exit the warning page and cancel the warning output signal at the same time.
Note: *when the Emergency Stop button is pressed down. Press Ⓒ key can exit the warning page and It allow to make other operation. But the warning output signal will not be cancelled.*

3.1 PARAMETERS OF "OPERATING PARAMETER" MENU

Parameter	Default value	Unit	Adjustable range	Remarks
Unload PRESS	Rated pressure +0.5	Bar	5.2~ (Rated pressure +0.5)	
Load PRESS	Rated pressure -0.5	Bar	4.5~ (Unloading pressure -0.5)	
Fan Start T	82	°C	Fan motor stop temperature ~160	
Fan Stop T	65	°C	0~ Fan motor start temperature	
Target Press.	Rated pressure	Bar	Rated pressure ~ Unload pressure	RS machine

- **Unload PRESS**
  When the compressor is running and the plant pressure is higher than the setting of unload pressure.The compressor will enter the unload status.
- **Load PRESS**
  When the compressor is running and the plant pressure is lower than the setting of load pressure, the compressor will enter the load status.
- **Fan Start/Stop T**
  The fan motor will start when the airend temperature reaches the value of "Fan motor start temperature"; The fan motor will stop when the airend temperature reaches the value of "Fan motor stop temperature".
- **Target Pressure (RS)**
  The value is the expected pressure of RS compressor running. When the plant pressure is less than the Target Pressure the invertor accelerates; When the plant pressure is more than the Target Pressure the invertor decelerates.

3.2 PARAMETERS OF "AUXILIARY PARAMETER" MENU

Parameter	Default value	Unit	Adjustable range	Remarks
Language	English	None	English/French/German/Italian/ Spanish	
Date	Current date	None		
Time	Current time	None		
P. unit	Bar	None	Mpa/Psi/Bar	
DI REM	OFF	None	OFF/ON	
Run-on	ON	None	OFF/ON	
Run-on time	180	sec	0~3600	
Slave IP	1	None	1~24	
BaudRate	9600	None	4800~38400	
Power Loss	OFF	None	OFF/ON	
Power Loss Time	5	sec	0~99	

- **Language**
  There are English/French/German/Italian/Spanish 5 languages interfaces for users to choose.
- **Date, time**
  The parameter can be used to change the real time.
- **P. unit**
  There are 3 pressure units for users to choose: MPa, Bar, and PSI.
- **DI REM**
  This parameter determines the compressor loading method. ON and OFF can enable remote load/unload function.
  OFF: Undoad valve controlled by the setting of load pressure and unload pressure.
  ON: Undoad valve controlled by the switch input terminal X09-DI6

- **Run-on, Run-on time**
  When the “Run-on” function is set to ON. The compressor’s sleep stop function will be activated. When the compressor begins to unload and keeps the unloaded status for the setting of “Run on Time”.The compressor will shut down automatically.
- **Slave IP**
  The address of the compressor as a slave on the sequence control status.
- **Baud Rate**
  The setting of RS485 communication port Baud rate
- **Power Loss on, Power Loss Time**
  When the setting of “Power loss on” is “ON”.The power-off restart function is activated.
  If the compressor is powered off while running. When it is powered on again. The compressor will automatically restart after the “Power Loss Time”.
  If the compressor is turned off normally before the power outage. It will not restart automatically after power on again.

3.3 PARAMETERS OF “MAINTENANCE RESET” MENU

Parameter	Default value	Unit	Adjustable range	Remarks
Air F Reset	Based on the actual service time	None	Resetting/reset already	Clear the actual service time of air filter
Oil F Reset	Based on the actual service time	None	Resetting/reset already	Clear the actual service time of oil filter
SEPA F Reset	Based on the actual service time	None	Resetting/reset already	Clear the actual service time of oil separation
Oil T Reset	Based on the actual service time	None	Resetting/reset already	Clear the actual time of oil change
GREA T Reset	Based on the actual service time	None	Resetting/reset already	Clear the actual time of motor maintenance
Cabinet F RES	Based on the actual service time	None	Resetting/reset already	Clear the actual service time of electric cabinet filter

3.4 PARAMETERS OF “COMPONENT ADJUSTMENT” MENU

Parameter	Default value	Unit	Adjustable range
SEPE PRESS Slope	0	None	-10~10
SEPE PRESS Zero	0	None	-10~10
Plant PRESS Slope	0	None	-10~10
Plant PRESS Zero	0	None	-10~10
Motor CUR A Slope	0	None	-10~10
Motor CUR A Zero	0	None	-10~10
Motor CUR B Slope	0	None	-10~10
Motor CUR B Zero	0	None	-10~10
Motor CUR C Slope	0	None	-10~10
Motor CUR C Zero	0	None	-10~10
Airend TEMP Slope	0	None	-10~10
Airend TEMP Zero	0	None	-10~10
Fan CUR Slope	0	None	-10~10
Fan CUR Zero	0	None	-10~10
AO Slope	0	None	-10~10
AO Zero	0	None	-10~10

Adjustment methods:

1. Zero calibration: It usually used for sensor calibration when the air compressor leaves the factory. When the analog input is 4mA which is zero value.But the display of it is not zero position. We can make the Zero calibration. The display value increased with setting.

2. Slope calibration: while the compressor running. It normally only needs to adjust the slope when the sensor is found to be incorrect. The display value increased with setting.

3.5 PARAMETERS OF "FACTORY PARAMETER" MENU

Parameter	Default value	Unit	Adjustable range	Remarks
Rate press	8	Bar	6.2-14	
Motor Current	0	A	0-1500	
Fan Current	0.0	A	0-200.0	
TEMP WARN	105	°C	0~stop temperature (F version)	
TEMP Trip	110	°C	Warning temperature ~120 (F version)	
PRESS Trip	Rated pressure +1.5	Bar	(Rated pressure +1.2)~ (Rated pressure +2.0)	
SEPA DP WARN	0.8	Bar	0~ Shutdown with oil separa-tion DP (<1)	
Running Time	Actual running time	h	0-999999	
Load Time	Actual loading time	h	0-999999	
Main CT Type	4000	:1	0-9999	
Fan CT Type	1000	:1	0-9999	
Pipe DP	0	Bar	0-1	
Fan Control	ON	None	ON/OFF	
Star/Delta Time	6	sec	0-99	
Air Filter Time	2000	h	0-9999	
Oil Filter Time	2000	h	0-9999	
Separator Time	4000	h	0-9999	
Oil Change Time	4000	h	0-9999	
Motor Greasing	2000	h	0-9999	
User Code	1111	None	0-9999	
Local/BLS	OFF	None	ON/OFF	
Factory Reset	OFF	None	ON/OFF	
AO Type	PI	None	P1/T/CT1	
Motor CUR IGNOR	3	sec	0-20	
Fan CUR IGNOR	2	sec	0-20	
BLS Load Time	1	sec	0-999	
BLS Unload Time	1	sec	0-999	
BLS Rotate Time	30	h	0-999	
BLS Start time	30	sec	0-999	
CUR Unbalance	20	%	5-90	
Loaded Valve	0	None	0/1	
SEPA DP Trip	1	Bar	Oil separation DP warning ~3	
Load Time	2	sec	0-30	
Soft Stop time	30	sec	0-300	
Cabinet Filter	1000	h	0-5000	
Min Start TEMP	1	°C	0-80	
Start Protect	0.8	Bar	0-1	
Motor Power	7.5	kW	7. 5/11/15/18. 5/22/30/37/45/55/75 90/110/132	
Main Kp	0.7		0^250. 0	

Parameter	Default value	Unit	Adjustable range	Remarks
Main Ti	0.8		0. T360. 0	
Freq. Max.	50.00	Hz	Calculated according to Motor Power	
Freq. Min.	26.00	Hz	Calculated according to Motor Power	
Accel T	10.0	S	0. T900. 0	
Decel T	10.0	S	0. T900. 0	
Fix FREQ ON	OFF	None	ON/OFF	
Fix FREQ	10.00	Hz	10. 00^90. 00	
Jump FREQ W	0.1	Hz	0~99. 99	
Jump RREQ 1	0.0	Hz	0~100. 0	
Jump FREQ 2	0.0	Hz	0~100. 0	

- **Rate PRESS**
  The setting of compressor rated pressure. The setting of unload pressure and load pressure and target pressure will be changed simultaneously when the rated pressure is modified. Please check the setting of unload pressure and load pressure and target pressure after completing the setting of this parameter.
- **Motor Current**
  The setting of Main motor current protection.Which related to the protection of main motor.
- **Fan Current**
  The setting of fan motor current protection. Which related to the protection of fan motor.
- **TEMP WARN**
  When the Airend discharge temperature is higher than the warning temperature. The controller will popup a warning message of “Warning High TEMP”.
- **TEMP Trip**
  When the Airend discharge temperature is higher than the trip temperature. The controller will popup a major fault message of “Fault High TEMP”. And the compressor will stop immediately.
- **PRESS Trip**
  When the plant pressure is higher than the trip pressure. The controller will popup a major fault message of “High Plant PRESS”. And the compressor will stop immediately.
- **SEPA DP WARN**
  When the oil separation DP is higher than the setting of oil separation DP warning. The controller will popup a warning message of “Warning Separator”
- **Running Time modification**
- **Load Time modification**
- **Main/Fan CT Type**
  Current transformer ratio of main motor and fan motor modification. Which according to the type of current transformer actually used.
- **Pipe DP**
  This parameter is related to the oil separation DP: oil separation DP = oil separator pressure-plant pressure-pipeline DP.
- **Fan Control**
  When the setting of this parameter is OFF. The fan motor control is invalidated; when it set to ON. The start/stop of the fan motor is according to the setting of “Fan Start T” and “Fan Stop T”.
- **Star/Delta Time**
  Time interval of switching the star start mode to the delta running mode of the main motor.
- **Air Filter Time**
  The service time setting of the air filter; when the actual service time of air filter reaches the setting. The compressor will popup a warning message of “Change Air Filter”.
- **Oil Filter Time**
  The service time setting of the oil filter; when the actual service time of oil filter reaches the setting. The compressor will popup a warning message of “Change Oil Filter”.
- **Separator Time**
  The service time setting of the oil separation; when the actual service time of oil separation reaches the setting. The compressor will popup a warning message of “Change Separator”.
- **Oil Change Time**
  The service time setting of the compressor oil; when the actual service time of compressor oil reaches the setting. The compressor will popup a warning message of “Change Oil”.
- **Motor Greasing**
  The service time setting of the motor grease; when the actual service time of motor grease reaches the setting. The compressor will popup a warning message of “Change Motor Greas.”.

- **User Code**
  The user access code setting
- **Local/BLS**
  Switch on/off the sequence control function.
- **Factory Reset**
  Restore the default values of factory parameter setting.
- **AO Type**
  Select the settings of analog output port. P1:discharge pressure (0~16Bar); T:Airend discharge temperature (-10~160°C); CT1:Main motor current A (0~setting of main motor rated current).
- **Motor CUR IGNOR**
  During the set time. The startup current of main motor will be ignored.To prevent false trigger current protection during startup phase.
- **Fan CUR IGNOR**
  During the set time. The startup current of fan motor will be ignored.To prevent false trigger current protection during startup phase.
- **BLS Load time, BLS Unload Time, BLS Rotate Time, BLS Start Time, CUR Unbalance**
  Please refer to section 4.3.1 for details.
- **Loaded Valve**
  Switch the Normally-open/normally-closed setting of unload valve.
- **SEPA DP Trip**
  When the oil separation DP is higher than the setting of oil separation DP trip. The controller will popup a fault message of "Fault Separator". And the compressor will stop.
- **Load Time**
  Delay time from Star-Delta period to unload valve energized.
- **Soft Stop Time**
  Delay time after the Stop key is pressed down.
- **Cabinet Filter**
  The service time setting of the electric cabinet filter; when the actual service time of electric cabinet filter reaches the setting. The compressor will popup a warning message of "Change Cabinet fil."
- **Min Start TEMP**
  Setting of the "Min Start TEMP". When the Airend discharge temperature is lower than the setting. The air compressor cannot be started.
- **Start Protect**
  When the oil separator pressure is higher than the setting of "Start Protect" pressure. The air compressor cannot be started.
- **Motor Power (RS)**
  The rated motor power of the unit. Which can be used to calculate max. and min, frequency of the unit in conjunction with rate pressure.
- **Main Kp (RS)**
  Proportional gain of invertor. For PID adjustment;
- **Main Ti (RS)**
  Integral time of invertor. For PID adjustment;
- **Freq. Max. (RS)**
  The maximum frequency setting of invertor;
- **Freq. Min. (RS)**
  The minimum frequency setting of invertor;
- **Accel T (RS)**
  Acceleration time of invertor from 0Hz to Freq.Max;
- **Decel T (RS)**
  Deceleration time of invertor from Freq.Max to 0Hz;
- **Fix FREQ ON (RS)**
  If the parameter set to "ON" the unit operates at a constant frequency without PID adjustment;
- **Fix FREQ (RS)**
  Constant operating frequency setting after the fixed frequency function is activated;
- **Jump FREQ W (RS)**
  The offset range of Jump FREQ point;
- **Jump FREQ 1 (RS)**
  Jump frequency point 1 of invertor;
- **Jump FREQ 2 (RS)**
  Jump frequency point 2 of invertor.

4.1 UNIT CONTROL

There are two control modes for start/stop of air compressor: local control and remote control. The air compressor's control mode can be selected via the status of X09-DI5 remote start/stop enable input.
The running status under any control mode can be stopped by pressing the Stop key on the controller's panel. Which can facilitate the emergency handling when a fault occurs.

4.1.1 LOCAL START/STOP CONTROL (FS MACHINE)

In the "Local control" mode. After the initialization of controller. Press Start key on the panel, the air compressor system will start up. And When the air compressor is running. Press Stop key on the panel, the air compressor will be DELAY STOP or UNLO.STOP after the set time.

1. Y—Δ Start:

Pressing the Start key . The unit starts up. The unit's startup process as follow: motor Y start status → delay time of "Factory Parameter- Star/Delta Time" → motor Δ running. Then the startup process of compressor ends. At starting phase all solenoid valves are deenergized. And the compressor starts with no load.

2. Running period:

When the motor startup enters the Δ status. After a period of delay time. The unload solenoid valve is energized. The air compressor runs loaded. And the plant pressure starts to rise. When the compressor plant pressure exceeds the setting of unload pressure. The unload solenoid valve is deenergized. And the air compressor runs unloaded. During the defined time. The plant pressure reduces to below the setting of load pressure. The loading solenoid valve will be energized again. And the compressor will run loaded. If the air compressor runs in unloaded status for a long period. For save energy. The "Run on" function can be activated.
The operating principle of the "Run on" function:
When the "Run on" function is activated. And the air compressor is unloaded. After the setting of "Run on Time". The air compressor will automatically shut down and enter the sleep status. When the plant pressure reduces to the setting of load pressure. The air compressor will automatically startup again.

3. Manually load/unload

When the compressor is unloaded. Press (manually load key). If the pressure is lower than the unload pressure setting. The unload valve is energized until the plant pressure is higher than the unload pressure setting. And it will return to the unloaded status. If the plant pressure is higher than the unload pressure when Press key. The manually load won't work.
When the compressor is loaded. Press (manually unload key) . The unload valve is deenergized until (manually load key) is pressed down and it will return to the loaded status.

4. Normal stop

Stop unloaded: when compressor stops in loaded status. The unload valve will deenergize. And after the delay time of "Factory paremeter-Soft Stop time". The motor contactor will deenergize.The main motor and the fan motor will stop running. Delayed stop: when it stops in unloaded status. After a delay of 20 seconds the motor contactor will deenergize. The main motor and the fan motor will stop running.

LOCAL START/STOP CONTROL(RS MACHINE)

In the "Local control" mode. After the initialization of controller. Press Start key on the panel. The air compressor system will start up. And When the air compressor is running. Press Stop key on the panel. The air compressor will be DELAY STOP or UNLO.STOP after the set time.

1. Start period:

Pressing the Start key . The invertor start to run with main motor. At starting phase all the solenoid valves are deenergized. And the compressor starts with no load.

2. Running period:

After the starting phase end. When the package plant pressure is lower than the unload pressure setting. The unload solenoid valve is energized and the air compressor is running in full load. And the plant pressure of the unit begins to rise. When the package plant pressure is higher than the unload pressure setting. The unload solenoid valve is deenergized again. The air compressor is running in offload and the invertor runs at the minimum frequency. When the air compressor runs in onload status. If the package plant pressure > the target pressure the invertor decelerates; If the package plant pressure < target pressure the invertor accelerates. If the air compressor runs in unload status for a long period. For save energy. The "Run on" function can be activated.
The operating principle of the "Run on" function: When the "Run on" function is activated. If the air compressor is running in offload for a period time, over the setting of "Run on Time". The air compressor will automatically shut down and switch to standby status. When the plant pressure down to the onload pressure setting. The air compressor will automatically startup again.

3. Manually load/unload

When the compressor is running in offload. Press < (manually load key). The unload solenoid energized and compressor switchs to load status in the case of plant pressure is less than unload pressure setting. But the compressor still be offload ff the plant pressure is higher than the unload pressure setting when Press < key.
When the compressor is running in onload. Press > (manually unload key). The unload solenoid valve is deenergized and compressor switch to offload status. To switch the compressor to onload press < (manually load key) again.

4. Normal stop

Press Stop button when compressor is running in onload status, the unload solenoid valve is deenergized and compresoor switchs to offload running. After escapes of “Factory paremeter-Soft Stop time”, the invertor stop Press Stop button when compressor is running in offload status, the compressor continuous to run in offload for 20 seconds then stop.

4.1.2 REMOTE START/STOP CONTROL

The different between remote start/stop control and local start/stop control is that the compressor start/stop by a remote Switch unless start button on controller panel. After the X09-DI5 input contact is closed. The remote start/stop control enable is activated. And the remote start signal will be triggered when the X09-DI2 input contact is closed. The remote stop signal will be triggered when the X09-DI2 input contact is opened.

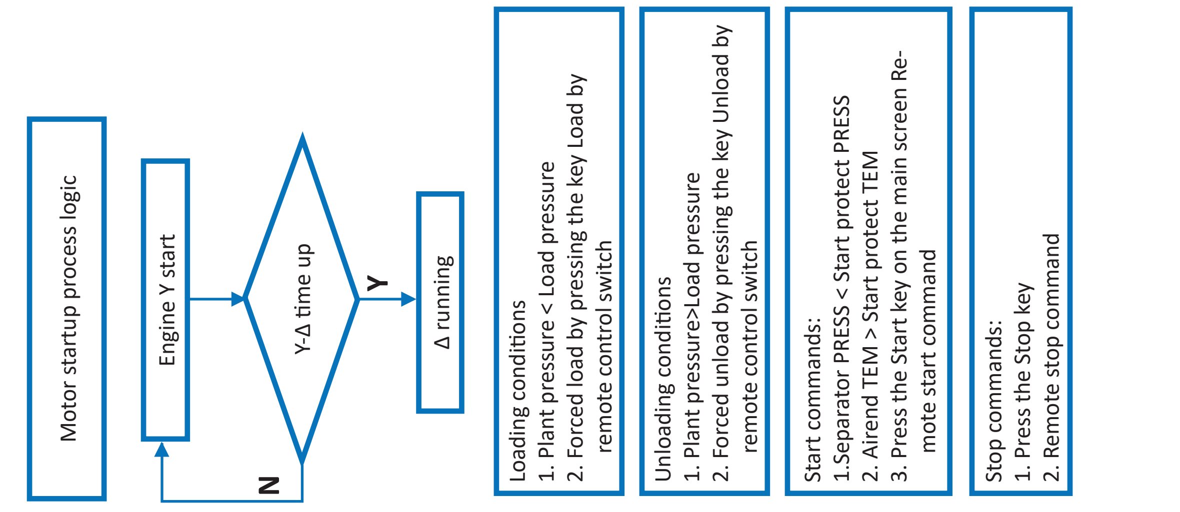
4.1.3 REMOTE LOAD/UNLOAD

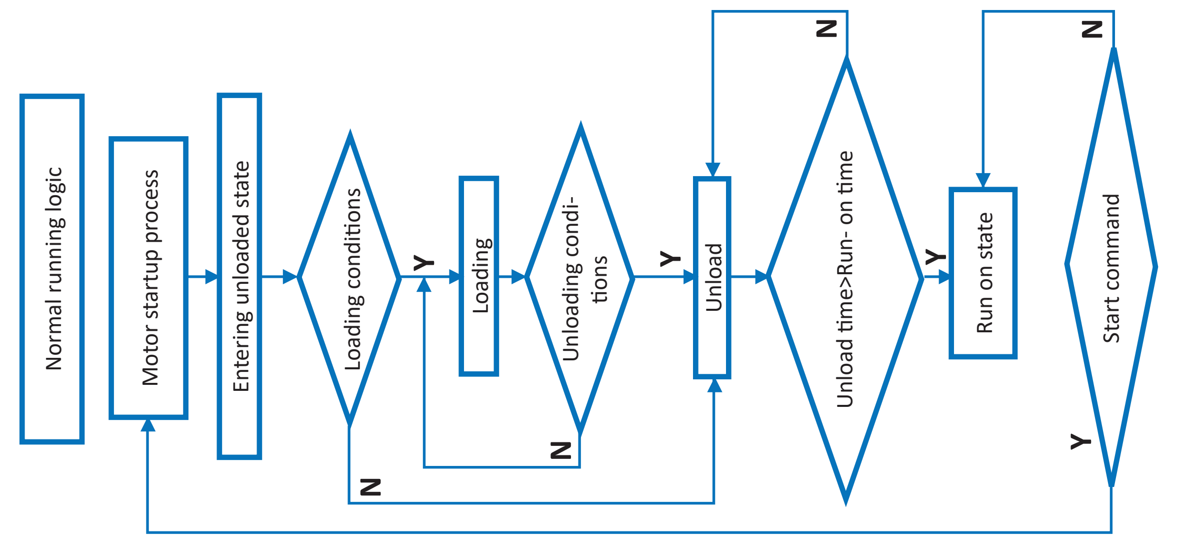
DI load: When parameter “Auxiliary Parameter-DI REM” enable is activated. And input contact X09-DI6 is closed. The air compressor runs in loaded status. And the running indicator is on. The controller displays “DI load”; when the plant pressure reaches the setting of unload pressure. DI load fails until the pressure reduces to the setting of unload pressure. When input contact X09-DI6 changes from “broken circuit” to “closed circuit” again. Then DI load is effective again. DI unload: When parameter “Auxiliary Parameter-DI REM” enable is activated. And input contact X09-DI6 is open. The air compressor runs in unloaded status. The running indicator is on. The controller displays “DI unload”.

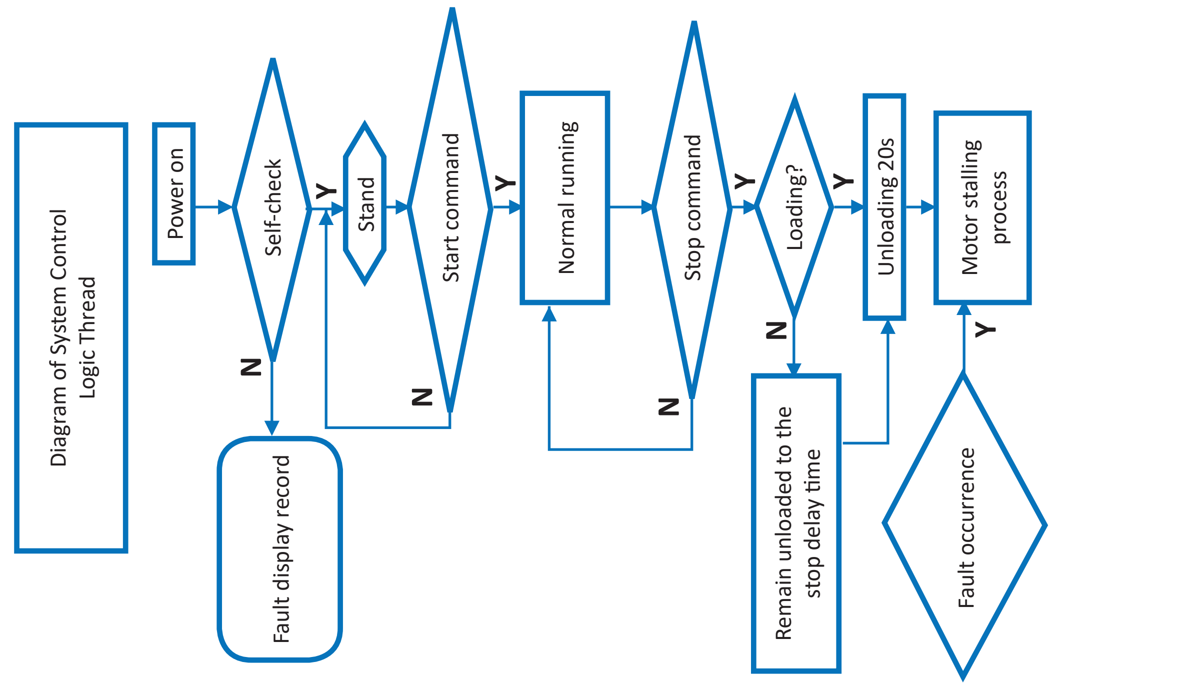
4.1.4 POWER-OFF RESTART

When the setting of “Auxiliary Parameter-Power loss on” to be “ON”.The air compressor will automatically recover to the running status before power-off after the time of “Auxiliary Parameter-Power loss time”. Upon being powered on after a power fault. If the air compressor is in running status before the power fault. The air compressor will be restarted automatically after passing the “Power loss time” upon being powered on again. If the air compressor is in stop status before the power fault. The air compressor will remain in stop status after being powered on again.

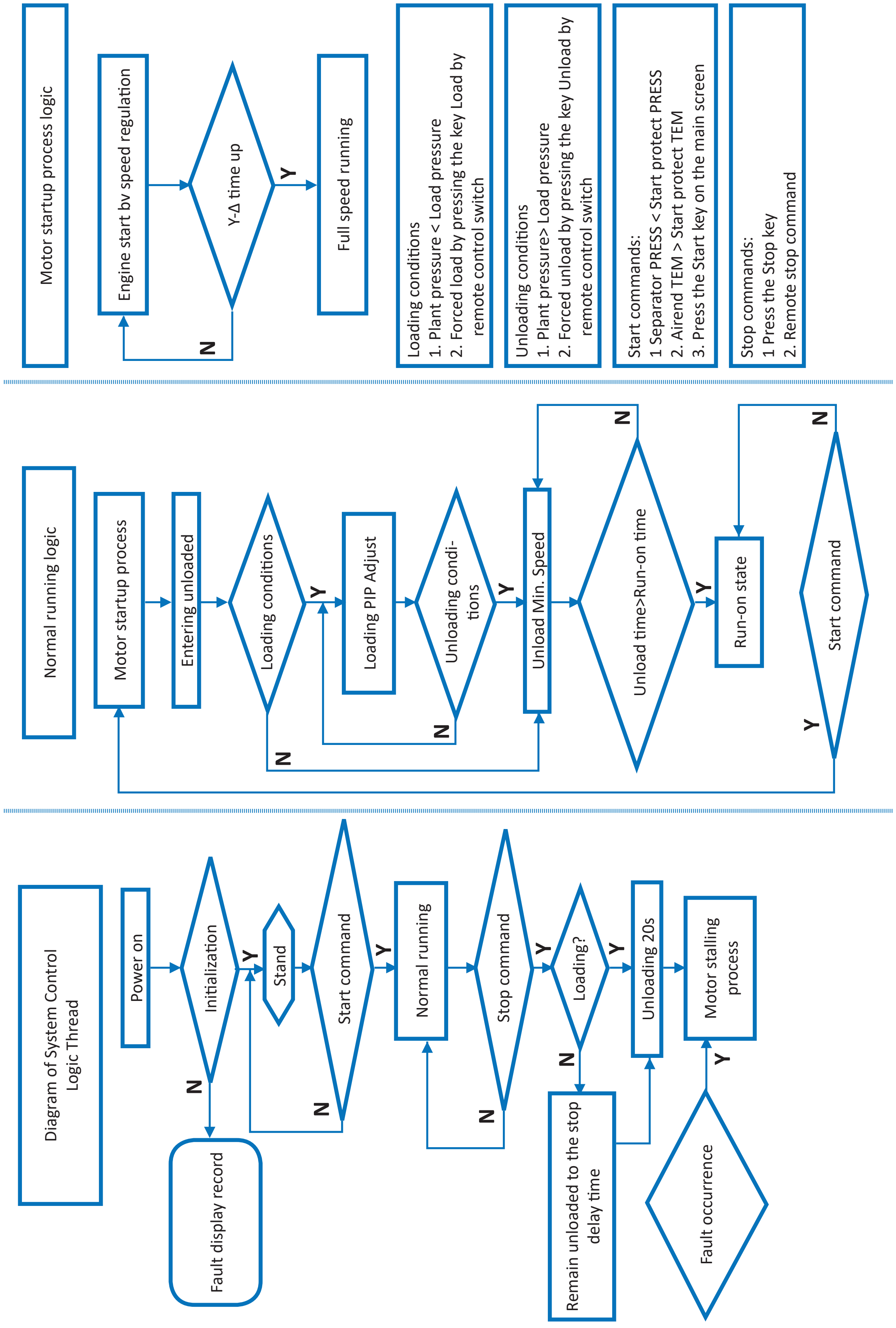
4.1.5 CONTROL LOGIC DIAGRAM (FS MACHINE)



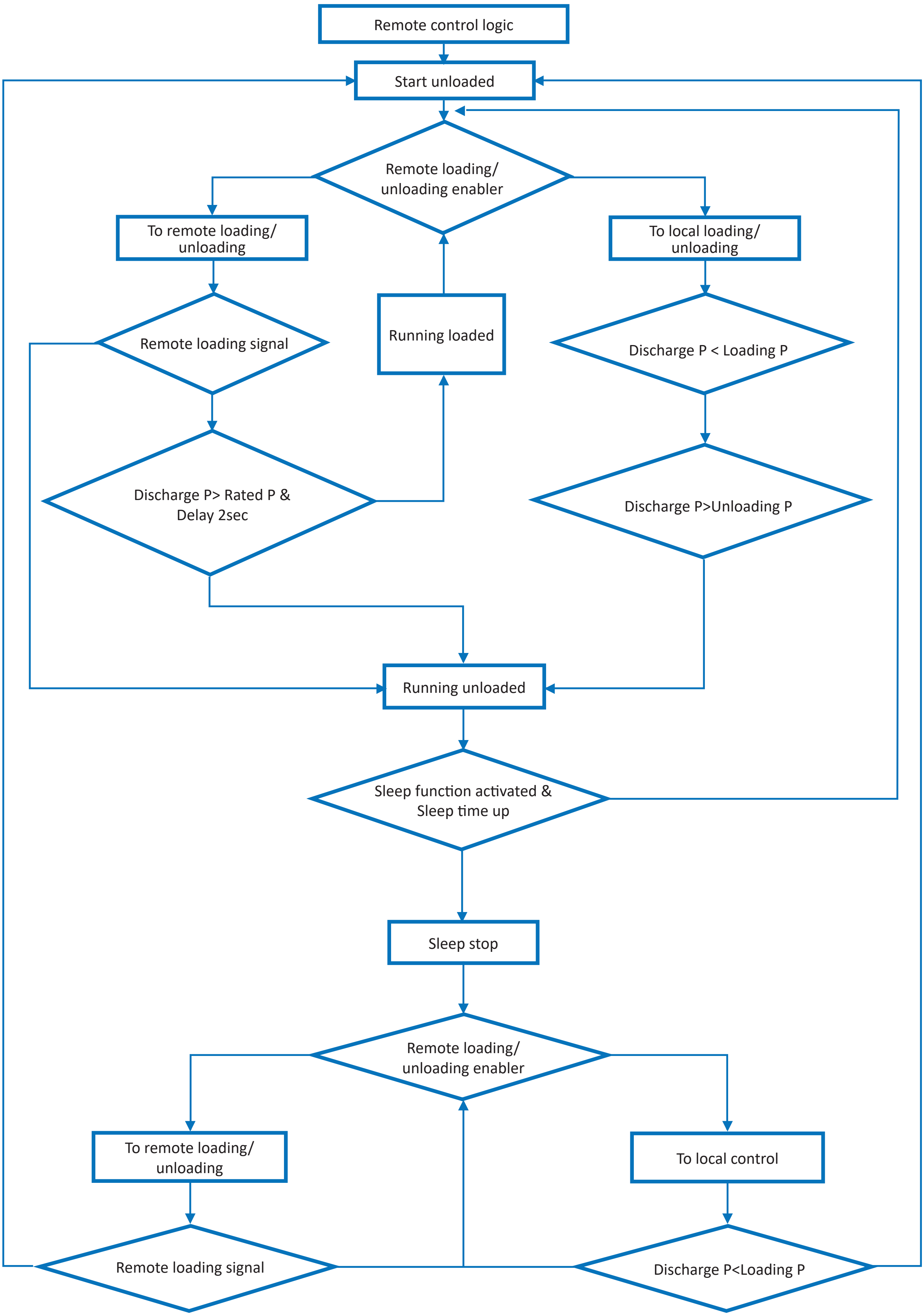




CONTROL LOGIC DIAGRAM (RS MACHINE)



REMOTE CONTROL LOGIC



4.2 FAN MOTOR CONTROL

When the “Factory Parameter- Fan Control” is turned off: The fan motor start or stop consistent with main motor. When the “Factory Parameter- Fan Control” is turned on: after main motor is started, the fan motor is controlled according to the Airend discharge temperature.
When the Airend discharge temperature is larger than the setting of “Operating Parameter-Fan start T”. The fan motor starts;
When the Airend discharge temperature reduces to below the setting of “Operating Parameter-Fan stop T”. The fan motor stops automatically.

4.3 SEQUENCE CONTROL (D VERSION MODIFIED)

It support up to 8 FS machines or 1 RS machine with up to 7 FS machines to form a sequence-control network via RS485 communication port. Each compressor sends and receives information from the network in sequence and processes the information. Every compressor can drop in and drop out the sequence-control network without influencing other compressors in the network.
The sequence-control network is simple, reliable, easily connectable and low-cost. Which enable multiple compressors to run stably under the working status. Also maintain the pipeline pressure within the set limit value and make the package’s work efficiency and improve the service life.

4.3.1 PARAMETERS SPECIFICATION OF SEQUENCE CONTROL

Firstly, connect the X10 communication ports of each compressor controller, which need to be entered to Sequence-control network, by using twisted shielded pair. Refer to wiring drawing.
***Note**: The GND terminal of X10 port of each compressor controller should be connected in the loop.*
The controller also need to be set in menu “Factory Parameter” as follow:

1. Sequential loading time setting (**BLS Load Time**): The time delay of the loading of each package in sequence when the pressure is lower than target;

2. Sequential unloading time setting (**BLS Unload Time**): The time delay of the unloading of each package in sequence when the pressure is higher than target;

3. Sequential start time setting (**BLS Start Time**): The time delay of the start up of each package in sequence;

4. The selection of control pressure shall be based on the highest running pressure of the packages in sequence. And the load/unload pressure setting of all packages should be the same.

4.3.2 OPERATING INSTRUCTIONS

1. Set the parameter “Factory Parameter-Local/BLS” to ON to enable the sequence control mode. You also need to set the parameter “Auxiliary Parameter -Slave IP” with different address numbers.

2. When text “F2 Confirm” appears on screen. Press key for acknowledgement; When screen displays “BLS Ready” means the compressor already dropped in sequence control network;

3. Press the Start key of any compressor in sequence network, the compressor will start immediately if there is no “Heavy start” alarm message on screen. If it fail to start next machine will take over the priority to start up. “Heavy start” means the compressor internal pressure is over 0.8bar start inhibition protection pressure when start up, the machine can start up automatically when internal pressure drop down to 0.8bar if plant pressure is still lower than load pressure setting +0.2Bar.

   The next compressor will start up in sequence with BLS Start Time delay when the plant pressure is still lower than load pressure setting +0.2Bar, the compressor which has min. running hours in network will start up in high priority. If compressors in network has same running hours, the compressor with a smaller Auxiliary Parameter-Slave IP address number will start up firstly. And so on.

   The next compressor will switch to unload in sequence with BLS Unload Time delay when the plant pressure is higher than unload pressure -0.2Bar, the compressor which has max. running hours in network will unload in high priority. If compressor in network has same running hours, the compressor with a bigger Auxiliary Parameter-Slave IP address number will unload firstly. And so on. The compressor will stop automatically when Run On Time escapes.

   When the plant pressure is lower than load pressure +0.2Bar, the compressors in network will start up in sequence with BLS Load Time delay. The compressor which has min. running hours in network will start up in high priority. If compressors in network has same running hours, the compressor with a smaller Auxiliary Parameter-Slave IP address number will start up firstly. And so on.

4. Pressing the Stop key or trip down with fault, the compressor will drop out the sequence network and “F2 Confirm” message appears on the screen. Press key to set the compressor back to sequence network again after troubleshooting.
   When one compressor drop out from network other compressors in same network will continue to work in sequence control w/o any interruption. To drop out all compressors in network you need to press Stop key one by one.

5. When a compressor trip down with fault or set back “Factory Parameter-Local/BLS” to OFF, the sequence control network will ignore this compressor and other compressors in same network will still works well.

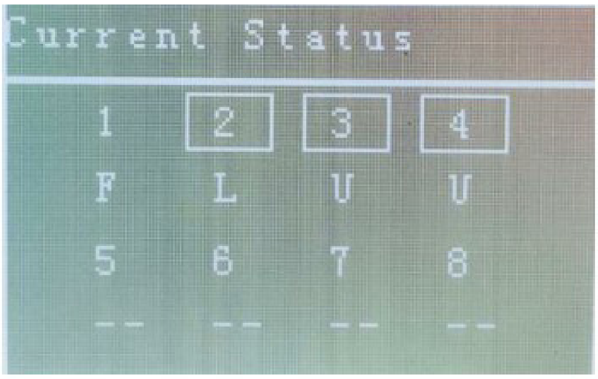
   When communication cable in network break off. The sequence control network is divided into two sub-networks. Compressors in each sub-network still can works in sequence.

6. The RS machine in network has the highest priority. The RS machine always drop in at first time when the plant pressure is lower than load pressure setting+0.2Bar and drop off at last when the plant pressure is higher than unload pressure setting-0.2Bar. Regardless RS machine’s running hours and slave IP address..

7. The control pressure of RS machine, which used to control invertor’s frequency, is the highest pressure of machines in network. Please make the same Load/unload pressure setting as other FS machines.

Notes:

1. *When the slave addresses number (Auxiliary Parameter-Slave IP) of different machines are set repeatedly, a warning message of “Warning BLS ID” will be displayed on screen;*
2. *When compressor power on at first time. You need to press > key to acknowledge the entering into sequence control network. A white window with slave address number will appear on the screen;*
3. *Pressing < key to enter the monitoring manu. It display the status of other compressors in same network. The number 1/2/3/4/5/6/7/8 on screen is the slave address numbers of each compressor in network. The letter under slave address number represents the status of each compressor (T: stop; R: start; L: load; U: unload; I: standby; F: fault).*
4. *When the Stop key is pressed or trip down with fault. The compressor will drop out the sequence control network, and the corresponding white window disappears on screen.*



5. *The compressor still can works independently when it drop out the sequence control network. The compressor don’t been controlled in sequence but it’s status still display on monitoring screen.*
6. *In sequence control. The pressure displayed on the top of screen is the maximum value of plant pressure in the network.*
7. *Message “F2 Confirm” means that you need to press > key to acknowledge entering into sequence control network; Message “BLS Ready” means that the compressor has dropped in the sequence control network and be ready in sequence control at any time; Message “BLS Run” means that the compressor is running in sequence control; Message “Communicated Control Monitoring” means that the compressor drop out the sequence control network and is running independently.*
8. *The Slave IP number of RS machine should be set to 1.*

5. Working status and fault status of air compressor

5.1 WORKING STATUS

1. Ready to start:
   The air compressor has no warning or fault.And the running indicator is off. The message “STA. Ready” is displayed in the status bar.

2. SD START:
   After the Start key is pressed. The air compressor star-delta start. And the running indicator will be on. The message “SD START” is displayed in the status bar. (FS)
   The Y-Δ switching time can be set in menu **Factory Parameter- Star/Delta Time**

Or After the Start key is pressed. The invertor start in offload. After load delay time escapes the running indicator is on and the message “SD START” is displayed on screen (RS)

3. LOAD:
   The air compressor is running in loaded status. And the running indicator is on. The message “LOAD” is displayed in the status bar. The unload valve is energized.

4. UNLOAD:
   The air compressor is running in unloaded status. And the running indicator is on. The message “UNLOAD” is displayed in the status bar. The unload valve is deenergized.

5. Local LOAD:
   Pressing the manually loading key < during compressor running. The air compressor will load. The running indicator will be on. And the message “Local Load” is displayed in the status bar.

6. Local UNLOAD:
   Pressing the manually unloading key > during running. The air compressor will unload out of pressure setting control. The run indicator will be on. And the message “Lo. UNLOAD” is displayed in the status bar.

7. Sleep running:
   When the air compressor unload for a long time until the time setting “Run on Time”. It will sleep stop and wait for the plant pressure drop to the unload pressure setting. And the running indicator will be on. The message “IDEL” is displayed in the status bar.

8. UNLOAD STOP:
   Pressing the Stop key while compressor is loading. The compressor will be in unloaded and stop after time setting **Factory Parameter- Soft Stop Time**. The message “UNLO.STOP” is displayed in the status bar. And the runningindicator will be off.

9. DELAY STOP:
   Pressing the Stop key while compressor is unloading. The compressor will be in delayed stop status. The message “UNLO. STOP” is displayed in the status bar. After 20 seconds it will stop and the running indicator will be off.

10. DI LOAD:
    When the parameter “**Auxiliary Parameter –DI REM**” enable is activated. And the X09-DI6 input contact has been closed. The air compressor will run in remote loaded status. And the running indicator will be on. The message “DI Load” is displayed in the status bar.

11. DI UNLOAD:
    When the parameter “**Auxiliary Parameter –DI REM**” enable is activated. And the X09-DI6 input contact has been opened. The air compressor will run in unloaded status out of pressure setting control.And the running indicator will be on. The message “DI Unload” is displayed in the status bar.

5.2 WARNING INFORMATION

The controller's LCD status bar will display corresponding warning messages alternately. Warning messages will not stop the compressor, simply prompting that the air compressor needs service.

List of light fault warning messages:

S/N	Warning message	Cause
1	Change Air Filter	The air filter service time exceeds the time setting (Factory Parameter-Air Filter Time)
2	Change Oil Filter	The oil filter service time exceeds the time setting (Factory Parameter-Oil Filter Time)
3	Change Separator	The oil separation service time exceeds the time setting (Factory Parameter-Separator Time)
4	Warning High TEMP	The Airend discharge temperature is higher than the setting of "Factory Parameter – TEMP WARN" and lower than the setting of "Factory Parameter –TEMP Trip"
5	Warning Separator	The oil separation DP is higher than the setting of "SEPA DP WARN" when the plant pressure is higher than 4.5 Bar
6	Change Oil	The service time of air compressor oil exceeds the time setting (**Factory Parameter- Oil Change Time**)
7	Change Motor Greasing	The motor grease maintenance time exceeds the time setting (F**actory Parameter- Motor Greasing**)
8	Warning BLS ID	There are repeated address setting of (**Auxiliary Parameter- Slave IP**)
9	Low Start TEMP	When the Airend discharge temperature is lower than the setting of "**Factory Parameter – MIN Start TEMP**". The air compressor can not start.
10	Heavy Startup	When the oil separator pressure is higher than the setting of "Factory Parameter – Start Protect". The air compressor can not start.
11	Change Cabinet filter	The service time of electric cabinet filter exceeds the time setting (**Factory Parameter- Cabinet Filter)**
12	Warning Air Filter	When the air filter is clogged. The contact of air filter DP switch will be open. Then the controller will give out message "Warning Air Filter"

5.3 FAULT SHUTDOWN

When a fault occurs and the air compressor automatically shuts down, the controller display interface indicates a warning message of the said major fault, and the air compressor cannot be re-started. After eliminating the fault, press C to exit the fault warning screen.

List of fault shutdown warning messages:

S/N	Fault	Cause
1	Fault High TEMP	The Airend discharge temperature is higher than the setting of "**Factory Parameter-TEMP Trip**"
2	Fault Cooling Fan	Feedback of fan breaker overload
3	Fault Direct ROT	5 seconds after compressor start. If the oil separator pressure is lower than 0.1Bar and shows no upward trend
4	High Plant PRESS	Plant pressure > "**Factory Parameter-PRESS Trip**"
5	Low load SEPA PRESS	In loaded status. The oil separator pressure is lower than 1.3 Bar for over a period of time
6	Low unload SEPA PRESS	In unloaded status. The oil separator pressure is lower than 0.6Bar over a period of time
7	Fault Sensor B1	The plant pressure sensor shorts circuit or disconnected
8	Fault Sensor B2	The oil separator pressure shorts circuit or disconnected
9	Fault Sensor R2	The Airend discharge temperature sensor shorts circuit or disconnected
10	Motor Locked-Rotor (FS)	After the startup process of compressor ends. When the motor current reaches 4 to 8 times the setting of "**Factory Parameter-Motor Current**" for 0.2s
11	Motor Stop Failed (FS)	The motor is still running in stop status
12	Fan Locked-Rotor	After the startup process of fan ends. When the fan current reaches 4 to 8 times the setting of "**Factory Parameter-Fan Current**" for 0.2s
13	Fan Stop Failed	The fan motor is still running in stop status
14	Fan Start Failed	The fan motor current is lower than 1A in start status
15	High SEPA PRESS	The oil separator pressure is higher than rated pressure+2 Bar
16	Fault Separator	In loaded status. The plant pressure > 4.5Bar. The oil separation DP > "**Factory Parameter-SEPA DP Trip**"
17	Motor Short Circuit (FS)	When the motor current reaches 8 times the setting of "**Factory Parameter-Motor Current**" for 0.2s
18	Fan Short Circuit	When the fan current reaches 8 times the setting of "**Factory Parameter-Fan Current**" for 0.2s
19	Motor Unbalance (FS)	After startup process. When the motor current imbalance is larger than the setting of "**Factory Parameter- CUR Unbalance**"
20	Fault EM-stop	The Emergency Stop button has been activated
21	Motor Overload (FS)	Motor overloaded (see table 5.3.1 **Inverse-time over-current protection**)
22	Fan Overload	Fan motor overloaded (see table 5.3.1 **Inverse-time over-current protection**)

S/N	Fault	Cause
23	Motor Phase A loss (FS)	After the compressor startup ends. A phase loss occurs
24	Motor Phase B loss (FS)	After the compressor startup ends. B phase loss occurs
25	Motor Phase C loss (FS)	After the compressor startup ends. C phase loss occurs
26	Fault Motor TEMP	The main motor PTC thermistor signal has been activated when the PTC resistor reaches 4K Ohm.
27	External fault	The contact of X09-DI7 has been activated
28	Drive fault (RS)	Invertor fault
29	Drive COM fault (RS)	The communication broken between Controller and Invertor.

5.3.1 INVERSE-TIME OVER-CURRENT PROTECTION

I Actual/I Set Time parameter	≥1.2	≥1.3	≥1.5	≥1.6	≥2.0	≥3.0
Duration time	60	48	24	24	5	1

6. Modbus-RTU communication settings

This controller provides X10 communication terminal and adopts MODBUS communication protocol. Users can make connection via RS485 from computer or PLC. Such as reading the working status of the controller.

Communnication Fromat: RTU mode
Character type: 8 digit data, no verification, 1 stop bit;
Check mode: CRC check;

List of UModbus Registers

Register (40×××)	Variable name	Read/Write	Range	Remarks
001	Control status	R		Refer the control status bits table as below
003	Plant pressure	R		Unit: 0.1 Bar
004	Oil separator pressure	R		Unit: 0.1 Bar
007	Airend temperature	R		Unit: °C
064	Running time (hour)	R		Unit: hour
065	Load time (hour)	R		Unit: hour
066	Running time (10.000 hours)	R		Unit: 10.000 hour
067	Load time (10.000 hours)	R		Unit: 10,000 hour
097	Pressure unit	R	0/1/2	0:MPa;l:PSI;2:Bar
098	Motor current A (FS)	R		Unit: A
099	Motor current B (FS)	R		Unit: A
100	Motor current C (FS)	R		Unit: A
112	Unload pressure	R	5.2~ (Rated pressure +0.5)	Unit: 0.1 Bar
113	Load pressure	R	4.5~ (Unload pressure -0.7)	Unit: 0.1 Bar
115	Y—Δ start time	R		Unit: Second
116	Run-on time	R		Unit: Second
121(BitO)	Power Loss On function	R	0/1	0:OFF;1:ON
122	Power Loss Time	R	10-120	Unit: Second
125	Fan motor current	R		Unit: 0.1 A
134	Slave IP	R	1-24	
254	Software version	R		
255	Warning codes	R		Refert the warning codes table as below
256-270	Fault codes record	R		Refert the fault codes table as below
280	COMM FREQ (RS)	R		Unit: 0.01HZ
281	Output FREQ (RS)	R		Unit: 0.01HZ
282	Output CUR (RS)	R		Unit: 0.1A
283	Output SPE (RS)	R		Unit: Rpm
284	Output VOL (RS)	R		Unit: 0.1 VAC
285	DC Bus VOL (RS)	R		Unit: VDC
286	Output POW (RS)	R		Unit: 0.01KW

Register (40×××)	Variable name	Read/Write	Range	Remarks
287	Drive TEMP (RS)	R		Unit: °C
288	Load rate (RS)	R		Unit: %
289	Target PRES (RS)	R		Unit: 0.1 Bar
290	Rate PRESS (RS)	R		Unit: 0.1 Bar
291	Motor Power RS)	R		Unit: 0.1KW
292	MAX FREQ (RS)	R		Unit: 0.01HZ
293	MIN FREQ (RS)	R		Unit: 0.01HZ

Control status bit	Definition	Read/Write	0	1	Remarks
Bit0	Reserve				
Bit1	Run / Stop	R	Stop	Run	
Bit2	Load / Unload	R	Unload	Load	
Bit3	Reserve				
Bit4	Reserve				
Bit5	Run-on	R	Non-sleep	In sleep	
Bit6	Fault	R	No fault	With fault	
Bit7	Warning	R	No warning	With warning	
Bit8	Reserve				

6. Modbus-RTU communication settings

The warning and fault codes table:

Fault No.	Fault prompt	Fault No.	Fault prompt
1	Change Air Filter	110	Fault High TEMP
2	Change Oil Filter	111	Fault Cooling Fan
3	Change Separator	112	Fault Direct ROT
4	Warning High TEMP	113	High Plant PRESS
5	Warning Separator	114	Low load SEPA PRESS
6	Change Oil	115	Low unload SEPA PRE.
7	Change Motor Greas	116	Fault Sensor B1
8	Warning BLS ID	117	Fault Sensor B2
9	Low Start TEMP	118	Fault Sensor R2
10	Heavy Startup	119	Motor Locked-Rotor
11	Change Cabinet filt	120	Motor Stop Failed
12	Warning Air Filter	121	Fan Locked-Rotor
122	Fan Stop Failed	131	Motor Overload
123	Fan Start Failed	132	Fan Overload
124	High SEPA PRESS	133	Motor Phase A loss
125	Fault Separator	134	Motor Phase B loss
126	Motor Short Circui	135	Motor Phase C loss
127	Fan Short Circuit	136	Fault Motor TEMP
128		137	External fault
129	Motor Unbalance	138	Invertor fault
130	Fault EM-stop	139	Invertor COM fault

Part No.: QX140501
Version: F (ECN: 1118958)
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